

Homework 6 - Scope and Parameters

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Problem 1. Runtime Stack (20 pts)

Consider the following block. Assume static scoping and call-by-value parameter passing.

```
1 { int x;  
2   int y;  
3   y := 2;  
4   { int f(int x) {  
5     if x==0 then {  
6       y := 1; }  
7     else {  
8       y := f(x-1)*4+x; }  
9     return y;  
10  }  
11  x := f(3);  
12 };  
13 };
```

1. [x: ?]
2. [x: ?, y: ?]
3. [x: ?, y: 2]
4. [x: ?, y: 2, f: <func>]

Lines 5-10 get skipped because f is being defined not ran.

11. [x: ?, y: 2, f: <func>] (Calls line 4) # x = f(3)

4. [x: 3] -> [x: ?, y: 2, f: <func>]

5. 3 != 0 so this is false, goes to else statement in 7

8. $[x: 3] \rightarrow [x: ?, y: 2, f: \langle \text{func} \rangle]$ (Calls line 4) $\# y: f(x-1) * 4 + x$

4. $[x: 2] \rightarrow [x: 3] \rightarrow [x: ?, y: 2, f: \langle \text{func} \rangle]$

5. $(2 \neq 0)$

So, this is false, goes to else statement in 7

8. $[x: 2] \rightarrow [x: 3] \rightarrow [x: ?, y: 2, f: \langle \text{func} \rangle]$ (Calls line 4) $\# y: f(x-1) * 4 + x$

4. $[x: 1] \rightarrow [x: 2] \rightarrow [x: 3] \rightarrow [x: ?, y: 2, f: \langle \text{func} \rangle]$

5. $(1 \neq 0)$

So, this is false, goes to else statement in 7

$[x: 1] \rightarrow [x: 2] \rightarrow [x: 3] \rightarrow [x: ?, y: 2, f: \langle \text{func} \rangle]$ (Calls line 4) $\# y: f(x-1) * 4 + x$

4. $[x: 0] \rightarrow [x: 1] \rightarrow [x: 2] \rightarrow [x: 3] \rightarrow [x: ?, y: 2, f: \langle \text{func} \rangle]$

5. $(x == 0)$

6. $[x: 0] \rightarrow [x: 1] \rightarrow [x: 2] \rightarrow [x: 3] \rightarrow [x: ?, y: 1, f: \langle \text{func} \rangle]$

9. $[x: 1] \rightarrow [x: 2] \rightarrow [x: 3] \rightarrow [x: ?, y: 1, f: \langle \text{func} \rangle]$ (return 1) $\# 1$ pop $[x: 0]$

9. $[x: 2] \rightarrow [x: 3] \rightarrow [x: ?, y: 5, f: \langle \text{func} \rangle]$ (return $1 * 4 + 1$) $\# f(0) * 4 + 1$ pop $x = 1$

9. $[x: 3] \rightarrow [x: ?, y: 22, f: \langle \text{func} \rangle]$ (return $5 * 4 + 2$) $\# f(1) * 4 + 2$ pop $x = 2$

9. $[x: ?, y: 91, f: \langle \text{func} \rangle]$ (return $22 * 4 + 3$) $\# f(2) * 4 + 3$ pop $x = 3$

11. $[x: 91, y: 91, f: \langle \text{func} \rangle]$ $\# f(3)$

Finish

Problem 2. Static and Dynamic Scope

Consider the following block. Assume call-by-value parameter passing.

```
1 {   int x;  
2     int y;  
3     int z;  
4     x := 1;  
5     y := 3;  
6     { int f(int y) { return x+y }  
7       int y;  
8       y := 5;  
9       { int g(int x) { return f(y) }  
10        { int y;  
11          y := 10;  
12          z := g(4);  
13        }  
14      }  
15    }  
16 }
```

a) (20 pts) Draw the runtime stack after each line executes under static scoping. What value assigned to z in line 12?

1. [x: ?]
2. [x: ?, y: ?]
3. [x: ?, y: ?, z: ?]
4. [x: 1, y: ?, z: ?]
5. [x: 1, y: 3, z: ?]
6. [x: 1, y: 3, z: ?, f: <func>]
7. [y: ?] → [x: 1, y: 3, z: ?, f: <func>]
8. [y: 5] → [x: 1, y: 3, z: ?, f: <func>]
9. [y: 5, g: <func>] → [x: 1, y: 3, z: ?, f: <func>]

10. [y: ?] → [y: 5, g: <func>] → [x: 1, y: 3, z: ?, f: <func>]

11. [y: 10] → [y: 5, g: <func>] → [x: 1, y: 3, z: ?, f: <func>] (next line is when split)

12. Call G(4) ← [x: 4]

9. f(5) (return f(y))

6. (return x + y) y is defined as 5 inside, x is defined as 1 in it's parent

Returns 5 + 1 = 6

9. (returns 6)

12. [y: 10] → [y: 5, g: <func>] → [x: 1, y: 3, z: 6, f: <func>]

Z would result in being 6 after line 12 finishes processing

b) (20 pts) Draw the runtime stack after each line executes under dynamic scoping. What value assigned to z in line 12?

1. [x: ?]

2. [x: ?, y: ?]

3. [x: ?, y: ?, z: ?]

4. [x: 1, y: ?, z: ?]

5. [x: 1, y: 3, z: ?]

6. [x: 1, y: 3, z: ?, f: <func>]

7. [y: ?] → [x: 1, y: 3, z: ?, f: <func>]

8. [y: 5] → [x: 1, y: 3, z: ?, f: <func>]

9. [y: 5, g: <func>] → [x: 1, y: 3, z: ?, f: <func>]

10. [y: ?] → [y: 5, g: <func>] → [x: 1, y: 3, z: ?, f: <func>]

11. [y: 10] → [y: 5, g: <func>] → [x: 1, y: 3, z: ?, f: <func>] (next line is when split)

12. G(4) ← [x: 4]

9. f(10) (return f(y))

6. (return x + y) y is defined as 10 inside, x is defined as 4 most recently

Returns $10 + 4 = 14$

9. (returns 14)

12. $[y: 10] \rightarrow [y: 5, g: \langle \text{func} \rangle] \rightarrow [x: 1, y: 3, z: 14, f: \langle \text{func} \rangle]$

Z would result in being 14 after line 12 finishes processing

Homework 6 - Scope and Parameters

Problem 3. Parameter Passing

Consider the following block. Assume dynamic scoping.

```
1 { int y;  
2   int z;  
3   y := 4;  
4   { int f(int b) {  
5       y := b+2;  
6       return (y-b)  
7   }  
8   int g(int x) {  
9       y := f(x+1)+5;  
10      z := f(x+y);  
11      return (z-1)  
12  }  
13  z := g(y*3);  
14  }  
15 }
```

- a) (20 pts) Draw the runtime stack after each line executes given that both parameters a and x are passed using Call-by-Name. What are the values of y and z after line 13 executes?

1. $[y: ?]$
2. $[y: ?, z: ?]$
3. $[y: 4, z: ?]$
4. $[y: 4, z: ?, f: \langle \text{func} \rangle]$

5-7 is the body of f

8. $[g: \langle \text{func} \rangle] \rightarrow [y: 4, z: ?, f: \langle \text{func} \rangle]$

8-12 is the body of g

13. calls g (y^*3)

8. $[x: y^*3] \rightarrow [g: \langle \text{func} \rangle] \rightarrow [y: 4, z: ?, f: \langle \text{func} \rangle]$

9. calls f ($y^*3 + 1$)

4. $[b: x + 1] \rightarrow [x: y^*3] \rightarrow [g: \langle \text{func} \rangle] \rightarrow [y: 4, z: ?, f: \langle \text{func} \rangle]$

5. $y = x + 1 + 2 \rightarrow y^*3 + 1 + 2 \rightarrow 4^*3 + 1 + 2 = 15$

5. $[b: x + 1] \rightarrow [x: y^*3] \rightarrow [g: \langle \text{func} \rangle] \rightarrow [y: 15, z: ?, f: \langle \text{func} \rangle]$

6. return y-b

6. $y - (y^*3 + 1) \rightarrow -15^*2 - 1 = -31$ is returned

9. $[x: y^*3] \rightarrow [g: \langle \text{func} \rangle] \rightarrow [y: -27, z: ?, f: \langle \text{func} \rangle] \# -31 + 4$

10. call f ($x + y$)

4. $[b: x+y] \rightarrow [x: y^*3] \rightarrow [g: \langle \text{func} \rangle] \rightarrow [y: -27, z: ?, f: \langle \text{func} \rangle]$

5. $y = b + 2$

5. $y = x + y + 2$

5. $y = 4 * -27 + 2$

5. $y = -108 + 2$

5. $[b: x+y] \rightarrow [x: y^*3] \rightarrow [g: \langle \text{func} \rangle] \rightarrow [y: -106, z: ?, f: \langle \text{func} \rangle]$

6. return y - b

6. return y - ($x + y$)

6. return -x

6. return -3 * y

6. return -3 * -106

10. [x: y*3] -> [g: <func>] -> [y: -106, z: 318, f: <func>]

11. return z - 1

11. return 318 - 1

13. [g: <func>] -> [y: -106, z: 317, f: <func>]

After line 13 **y = -106**, and **z = 317**

b) (20 pts) Draw the runtime stack after each line executes given that both parameters a and x are passed using Call-by-Need. What are the values of y and z after line 13 executes?

1. [y: ?]
2. [y: ?, z: ?]
3. [y: 4, z: ?]
4. [y: 4, z: ?, f: <func>]

5-7 is the body of f

8. [g: <func>] → [y: 4, z: ?, f: <func>]

8-12 is the body of g

14. Calls g ($y * 3$) = $g(4 * 3)$ = g (12)

8. [x: 12] → [g: <func>] → [y: 4, z: ?, f: <func>]

9. Calls f(x+1) = f(12+1) = f(13)

4. [b: 13] → [x: 12] → [g: <func>] → [y: 4, z: ?, f: <func>]

5. $y = b + 2$

5. $y = 15$

5. [b: 13] → [x: 12] → [g: <func>] → [y: 15, z: ?, f: <func>]

6. return y-b

6. return $15 - 13 = 2$

9. $y = 2 + 5$

9. [x: 12] → [g: <func>] → [y: 7, z: ?, f: <func>]

10. Calls f(x+y)

10. Calls f(12 + 7) = f(19)

4. [b: 19] → [x: 12] → [g: <func>] → [y: 7, z: ?, f: <func>]

5. $y = b + 2$

5. $y = 19 + 2 = 21$

5. [b: 19] → [x: 12] → [g: <func>] → [y: 21, z: ?, f: <func>]

6. return y - b

6. return $21 - 19 = 2$

6. [x: 12] → [g: <func>] → [y: 21, z: ?, f: <func>]

10. [x: 12] → [g: <func>] → [y: 21, z: 2, f: <func>]

11. return z-1

11. return $2 - 1 = 1$

11. [g: <func>] -> [y: 21, z: 1, f: <func>]

```
1 {   int y;
2     int z;
3     y := 4;
4     { int f(int b) {
5         y := b+2;
6         return (y-b)
7     }
8     int g(int x) {
9         y := f(x+1)+5;
10        z := f(x+y);
11        return (z-1)
12    }
13    z := g(y*3);
14 }
15 }
```